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21 MAY 1953
Superseding
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12 November 1952

MILITARY SPECIFICATION

RELAY, ARMATURE, EXCITER-CONTROL, 115/200 VOLT, 3-PHASE 4-WIRE 400-CYCLE, TYPE H-1

This specification has been approved by the Department of the Air Force and by the Navy Bureau of Aeronautics.

1. SCOPE

1.1 This specification covers one type of intermittent-duty coil latch-type relay, designated Type H-1.

2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS, AND PUBLICATIONS

2.1 The following specifications, standard, drawing, and publication, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-D-5028	Drawing and Data Lists; Preparation of (For Engines, Accessories and Other Auxiliary Equipment)
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-M-7911	Marking, Identification of Aeronautical Equipment, Assemblies, and Parts
MIL-P-6064	Packaging of Lightweight Aircraft Accessories
MIL-S-7742	Screw Threads, Standard, Aeronautical

STANDARDS

MIL-STD-129	Marking of Shipments

DRAWINGS

Air Force-Navy Aeronautical Standard Drawing

AN3102	Connector; Receptacle - Box Mounting

PUBLICATIONS

Air Force-Navy Aeronautical Bulletin

No. 143	Specifications and Standards; Use of

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(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Materials.-

3.1.1 Dissimilar metals.- Unless suitably protected, dissimilar metals which tend to set up a galvanic action in the presence of moisture or salt spray shall not contact each other.

3.1.2 Protective treatment.- When materials are used in the construction of the exciter-control relay that are subject to corrosion in salt air or other atmospheric conditions likely to occur during service usage, they shall be protected against such corrosion in a manner that will in no way prevent conformance with the requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of atmospheric conditions shall be avoided.

3.1.3 Selection of materials.- Specifications and standards for all materials, parts, and Government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with ANA Bulletin No. 143, except as provided in the following paragraph.

3.1.3.1 Standard parts.- Standard parts (MS, AN, or JAN) shall be used wherever they are suitable for the purpose, and shall be identified on the drawing by their part numbers. Commercial utility parts such as screws, bolts, nuts, cotter pins, etc, may be used, provided they possess suitable properties and are replaceable by the standard parts (MS, AN, or JAN) without alteration, and provided the corresponding standard part numbers are referenced in the parts list and, if practicable, on the contractor's drawings. In the event there is no suitable corresponding standard part in effect on date of invitation for bids, commercial parts may be used provided they conform to all requirements of this specification.

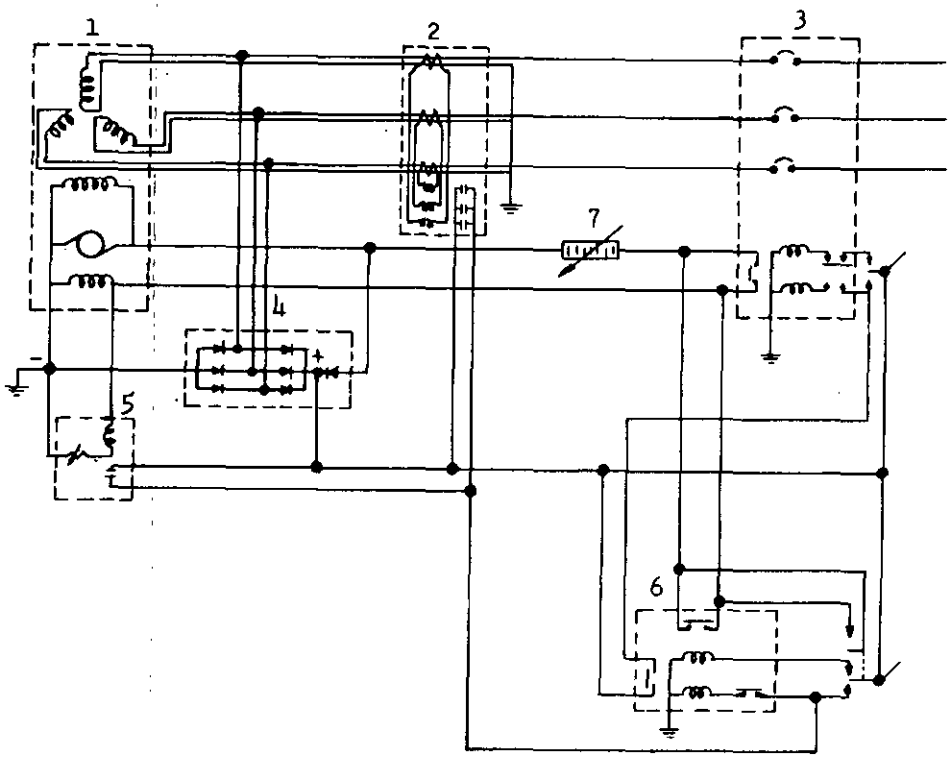
3.2 Design.- The relay shall be designed as a latch-type relay with a closing coil and a tripping coil with contacts to open or close the exciter field circuit in conjunction with the operation of the differential-current relay, exciter-protection relay, and limiting-switch relay. Sufficient contacts for the operation of the differential-current relay, exciter-protection relay, and limiting-switch relay shall be provided on the exciter-control relay. The close coil and tripping coil shall be designed for operation from a 28V d-c power source and controlled by the limiting switch, differential-current relay, or the exciter-protection relay. The exciter-control relay contacts shall be designed to open and close the exciter field circuit, trip the line circuit breaker, and provide other interlocking control circuits.

3.2.1 A typical circuit diagram for an alternator, circuit breaker, differential-current relay, exciter-protection relay, and exciter-control relay is shown on figure 1. The various coils and contacts shall be connected to a suitable AN connector as shown on figure 2.

3.3 Construction.- The relay shall be so constructed that no parts will work loose in service. It shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service.

3.3.1 The relay shall be so constructed that adjustment and repairs can be easily made by the personnel of operating units and overhaul bases.

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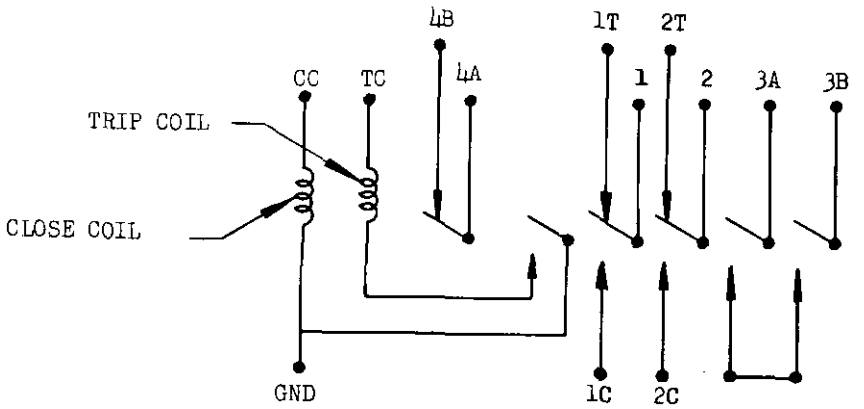
THIS CIRCUIT SHOWS TYPICAL CONNECTIONS ONLY AND DOES NOT NECESSARILY REPRESENT THE EXACT EMPLOYMENT INTENDED FOR THE EXCITER-CONTROL RELAY

LEGEND

1. ALTERNATOR
2. DIFFERENTIAL-CURRENT RELAY
3. CIRCUIT BREAKER
4. 21-AMPERE TRANSFORMER RECTIFIER
5. EXCITER-PROTECTION RELAY
6. EXCITER-CONTROL RELAY
7. CARBON PILE FOR VOLTAGE REGULATOR

FIGURE 1. Relay

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CONTACTS 3A, 3B TO BE DOUBLE BREAK

CONTACTS 4A, 4B TO BE ACTUATED DIRECTLY BY THE TRIPPING ELEMENT

THIS DIAGRAM ASSUMES THAT THE LAST COIL ENERGIZED WAS THE TRIP COIL

FIGURE 2. Coils and contacts

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3.4 Interchangeability.- All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part numbers shall be governed by the drawing number requirements of Specification MIL-D-5028.

3.5 Tripping and closing coils.-

3.5.1 Operation of closing and tripping coils.- There shall be two coils (tripping and closing) on the relay suitable for operation on 28V d-c power. The closing coil shall be suitable for closing the contacts and compressing or extending a spring of suitable size and strength necessary to force the contact arm to the other extreme position. When the closing coil has brought the contact arm to an extreme position, a latch mechanically connected to the tripping coil shall prevent the spring from moving the contact arm until the tripping coil has been energized.

3.5.2 Rating of closing and tripping coils.- The closing coil shall operate at any voltage between 12 and 30V d-c and shall be designed to withstand any voltage between these limits for 1 minute. The tripping coil shall be an intermittent-duty coil and shall satisfactorily operate on any direct-current voltage between 12 and 30V.

3.6 Contacts.- There shall be two single-pole double-throw contacts and one single-throw double-break contact as shown on figure 2, all suitable for operation on 200V ac. All single-throw contacts shall close when the close coil is energized. There shall be a single-break contact to open the trip-coil circuit when it is energized. An additional contact shall be provided to operate in conjunction with the tripping element. This contact shall close within 0.008 second after any voltage between 12 and 30V is applied to the tripping coil. The double break contacts 3A and 3B are intended to be used in the exciter field circuit.

3.7 Insulation.- All electrical conductors on the relay shall be insulated from the frame of the relay. The coil windings shall be capable of withstanding 1,000V rms, 60 cycles, between the coil terminals and the relay frame. The contacts shall be capable of withstanding 1,500V rms, 60 cycles, between the contact terminals and relay frame. Air gaps between live parts and the relay frame shall be at least 1/8 inch. Creepage distances of at least 1/4 inch shall be maintained, except for parts in operating-coil circuits where 1/8 inch is permissible, and in AN connectors.

3.8 Size.- The relay shall have maximum over-all dimensions and mounting hole dimensions as shown on figure 3.

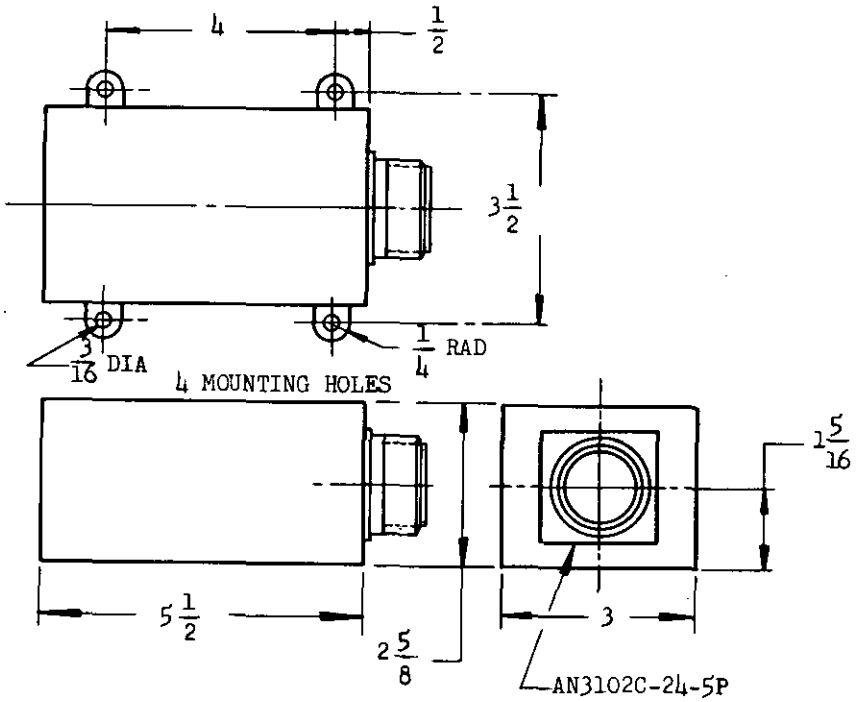
3.9 Base and cover.- The relay and AN connector shall be fastened to the base of the relay. A removable cover shall be provided to minimize the effects of moisture, sand, and salt spray.

3.10 Terminals.- The terminal arrangement and marking shall be as shown on figure 3.

3.11 Weight.- The relay shall not weigh more than 1-1/2 pounds.

3.12 Screw threads.- Unless otherwise specified, the threads of all machine screws, 0.060 inch, or larger, in diameter shall conform to Specification MIL-S-7742.

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TERMINAL CONNECTIONS	
PIN	RELAY
A	CC
B	TC
C	GND
D	1C
E	1
F	3A
G	1T
H	3B
J	4A
K	4B
L	2C
M	
N	2T
P	
R	
S	2

DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED. TOLERANCES: FRACTIONS ±1/32

FIGURE 3. Mounting holes and envelopes

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3.13 Performance.- The relay shall be capable of operating under the following conditions:

(a) Temperature: A minimum ambient temperature of -70°C at all altitudes. A maximum ambient temperature of +60°C at sea level varying uniformly to -40°C at 40,000 feet altitude, and -40°C from 40,000 to 50,000 feet altitude

(b) Altitude: A pressure-altitude range from sea level to 50,000 feet altitude.

(c) Humidity: A humidity ranging from saturation at tropical sea level +40°C, to saturation at 50,000 feet altitude, -70°C.

(d) Sand and dust: Exposure to airborne sand particles as encountered in desert operation.

(e) Salt spray: Exposure to atmosphere containing salt-laden moisture.

(f) Vibration: Conditions of linear and torsional vibration incident to normal continuous use in aircraft. Any required shock-mounting provision shall be included as an integral part of the relay.

(g) Fungi: Moist fungi encountered in tropical and semitropical climates.

(h) Life: The relay shall be capable of withstanding 5,000 cycles of operation under normal conditions.

3.14 Identification of product.-

3.14.1 Nameplate.- The relay shall be marked for identification in accordance with Specification MIL-M-7911. The nameplate shall contain the following information:

RELAY, ARMATURE EXCITER CONTROL
Type H-1
Specification MIL-R-7649A(ASG)
Stock No. (USAF or Navy, as applicable)
Manufacturer's Part No.
Manufacturer's Serial No.
Contract or Order No.
Manufacturer's name or trade-mark
US Property

3.14.2 Use of AN or MIL designations.- AN or MIL designations shall not be applied to a product, except for Qualification test samples, nor referred to in correspondence, until notice of approval has been received from the activity responsible for qualification, or from the Aeronautical Standards Group.

3.15 Workmanship.- The relay, including all parts and accessories, shall be fabricated and finished in a workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, burrs, and sharp edges; thoroughness of soldering and wiring; alinement of parts and tightness of assembly screws.

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4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Classification of tests.- The inspection and testing of the relay shall be classified as follows:

(a) Qualification tests: Qualification tests are those tests accomplished on samples submitted for qualification as a satisfactory product.

(b) Inspection tests: Inspection tests are those tests accomplished on relays manufactured and submitted for acceptance under contract.

4.2 Test conditions.-

4.2.1 Standard atmospheric conditions.- Whenever the pressure and temperature existing at the time of the test are not definitely specified, it is understood that the test is to be made at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature (approximately +25°C). When tests are made with atmospheric pressure or room temperature differing materially from the specified values, proper allowance shall be made for the difference from the conditions specified.

4.3 Qualification tests.-

4.3.1 Prior qualification.- Unless otherwise specified by the procuring activity, relays which have not previously passed a Qualification test, or which have passed the Qualification tests and have been modified in any manner shall satisfactorily pass the Qualification tests prior to the acceptance of any relay.

4.3.2 Sampling instructions.- Qualification test samples shall consist of two relays. Samples shall be identified as required and forwarded to the activity responsible for qualification, designated in the letter of authorization from the qualifying activity. (See paragraph 6.2.)

4.3.3 Tests.- The Qualification tests of relays shall consist of all the tests specified under Inspection tests and, in addition, those described herein.

4.3.3.1 Unless otherwise specified herein, environmental tests shall be conducted in accordance with Specification MIL-E-5272 and in the following order:

Exciter-control relay No. 1	Exciter-control relay No. 2
(a) High temperature	(a) Life
(b) Low temperature	(b) Vibration
(c) Humidity	(c) Salt spray
(d) Fungus	(d) Sand and dust
(e) Explosion-proof	

4.3.3.1.1 High temperature; Procedure I.- During the final hour of the test, the relay shall be subjected to and shall satisfactorily withstand the Performance test specified herein.

4.3.3.1.2 Low temperature: Procedure II.- During the final hour of the test, the relay shall be subjected to and satisfactorily withstand the Performance test specified herein.

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4.3.3.1.3 Humidity: Procedure I.- At the end of the test, the relay shall show no sign of corrosion of metal parts. The relay shall then be subjected to and shall satisfactorily withstand the Performance and Insulation tests specified herein. In the latter tests, potentials of 750V shall be used in lieu of 1,500V.

4.3.3.1.4 Vibration: Procedure I.- The relay shall be vibrated with the close coil closed half of the time, and with the trip coil closed the remainder of the time. All contacts in the closed position shall remain closed and all contacts in the open position shall remain open. At the end of the test, the relay shall be subjected to and shall satisfactorily withstand the Performance test specified herein.

4.3.3.1.5 Salt spray: Procedure I.- At the end of the 50-hour period, the relay shall show no signs of corrosion of metal parts and shall be subjected to and satisfactorily withstand the Performance test specified herein.

4.3.3.1.6 Sand and dust: Procedure I.- If at the end of the test there is a collection of dust visible inside the case, the relay shall be subjected to and satisfactorily withstand the Performance test specified herein.

4.3.3.1.7 Fungus: Procedure I.- At the end of the test period, the relay shall be inspected and shall be subjected to and shall satisfactorily withstand the Performance and Insulation tests specified herein. The test potentials for the Insulation test shall be 500 and 750V in lieu of 1,000 and 1,500V, respectively.

4.3.3.1.8 Explosion-proof: Procedure I.- When the relay is operated during this test no explosion shall occur.

4.3.3.2 Life.- The relay shall be subjected to 5,000 opening and closing operations at a frequency of approximately 4 cpm, using a control voltage of 18V dc. The operating cycle shall be 10 seconds with the close coil closed, and 5 seconds with the close coil open. The double-break contacts shall be connected to a 10-amp circuit at 28V dc. Operation shall be satisfactory during the test. At the end of the test, the relay shall be subjected to and shall satisfactorily withstand the Performance test specified herein.

4.4 Inspection tests.- The Inspection tests shall consist of Individual tests and Sampling tests.

4.4.1 Individual tests.- Each relay shall be subjected to the following tests. In addition, each relay shall be subjected to any other test specified herein which the Inspector considers necessary to determine compliance with the requirements of this specification.

4.4.1.1 Examination of product.- Each relay shall be inspected to determine compliance with the requirements specified herein with respect to material, workmanship, and marking.

4.4.1.2 Performance.- A potential of 30V dc shall be applied to the close coil and maintained for 1 minute. The relay shall successfully operate, and the coil shall not be damaged. The close coil shall also operate the relay successfully on a potential of 12V dc. Potentials of 30 and 12V dc shall be applied to the tripping coil, and in each case the relay shall operate properly.

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4.4.1.3 Insulation tests.- A potential of 1,000V rms, 60 cycles, shall be applied between a coil terminal and relay frame for 1 minute. No insulation failure shall result. The relay shall successfully withstand 1,500V rms, 60 cycles, for 1 minute between the relay frame and contact terminal, except the contact, which is connected in the trip-coil circuit.

4.4.2 Sampling tests.- Five relays which have been manufactured under essentially the same conditions shall be selected at random from each lot of 500 or fraction thereof on the order and subjected to the following tests.

4.4.2.1 Test on operating time.- Oscillographic tests shall be conducted to determine the closing time of the contacts actuated by the tripping coil after the application of 12V dc to the tripping coil. The relay shall be tested three times, and the average closing time shall not exceed 0.008 second.

4.4.3 Rejection and retest.- When Sampling tests are specified on a number of items that are selected as representative of a certain lot, and one or more of this number fails to meet the specified test(s), acceptance of all items shall be withheld until the extent and cause of failure is determined. For operational reasons, Individual tests may be continued pending investigation of a Sampling test failure, but the final acceptance of the product is contingent upon the Inspector's decision regarding the over-all conformance of the product to specification requirements. If investigation indicates that the defects may exist on items previously accepted, full particulars concerning the defect(s) found, including recommendations for correction, will be furnished to the procuring activity.

4.5 All parts, specimens, or assemblies destroyed in making tests required by this specification, to determine compliance with the specification, shall be in addition to the quantity specified in the contract or purchase order and shall be furnished without increasing the cost of the contract or order.

5. PREPARATION FOR DELIVERY

5.1 Application.- The requirements specified herein apply only to direct purchases by or direct shipments to the Government.

5.2 Packaging and packing.-

5.2.1 Domestic and overseas packing.- Relays shall be preserved, packaged, and packed in accordance with Specification MIL-P-6064, group X, type II, class I, size 2 container.

5.3 Marking of shipments.- Interior packages and exterior shipping containers shall be marked in accordance with Standard MIL-STD-129. The nomenclature shall be as follows: Relay, Armature Exciter-Control 115/200 Volt, 3-Phase, 4-Wire, 400-Cycle, Type H-1, (Manufacturer's Part No. *), Specification MIL-R-7649A(ASG).

* Applicable data to be entered by the contractor.

6. NOTES

6.1 Intended use.- The Type H-1 exciter-control relay covered by this specification is an intermittent-duty coil-latch type relay for use on 115/200V, three-phase, four-wire systems in which the coil power is 28V dc, and for use with an alternator in accordance with Specification 32509.

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6.2 Provisions for Qualification tests.- In the procurement of products requiring qualification, the right is reserved to reject bids on products that have not been subjected to the required tests and found satisfactory for inclusion on the Military Qualified Products List. The attention of suppliers is called to this requirement, and manufacturers are urged to communicate with the Commanding General, Wright Air Development Center, Wright-Patterson Air Force Base, Ohio, the activity responsible for qualification, with a copy to the Bureau of Aeronautics, Navy Department, Washington 25, D. C., and arrange to have the products that they propose to offer to the Navy or the Air Force, tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products covered by this specification may be obtained from the above designated activity responsible for qualification.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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